ReviewSense AI

AI Engineer Bootcamp Project

From 50,000+ Reviews to 1 Smart Recommendation

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ReviewSense AI

The Goal

From 50,000+ Reviews to 1 Smart Recommendation

Sentiment Analysis

(DistilBERT)

Product Clustering

(MiniLM + KMeans)

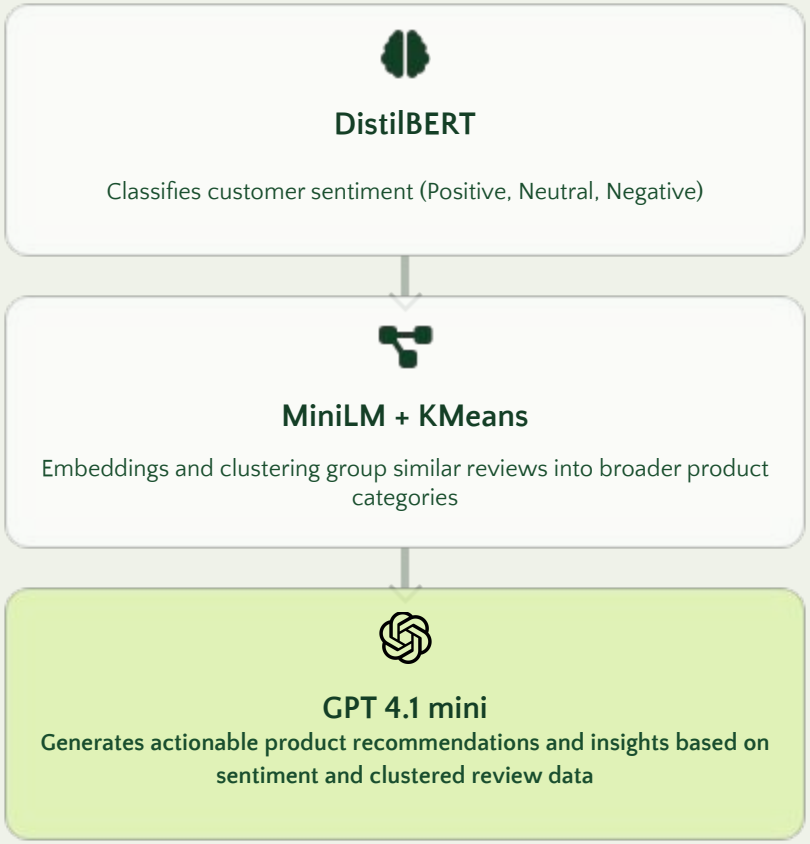
AI-Generated Buying Recommendations

(Generative AI)

"Stop reading thousands of reviews. Let AI summarize them for you."

Project Overview

Pipeline Architecture



Methodology

- Combines multiple AI models into one end-to-end review analysis pipeline
- Structured insights are used by Generative AI models to create short recommendation articles
- **Output:** Top products, common complaints, and smart buying recommendations

METHODS & DATASET

Overview

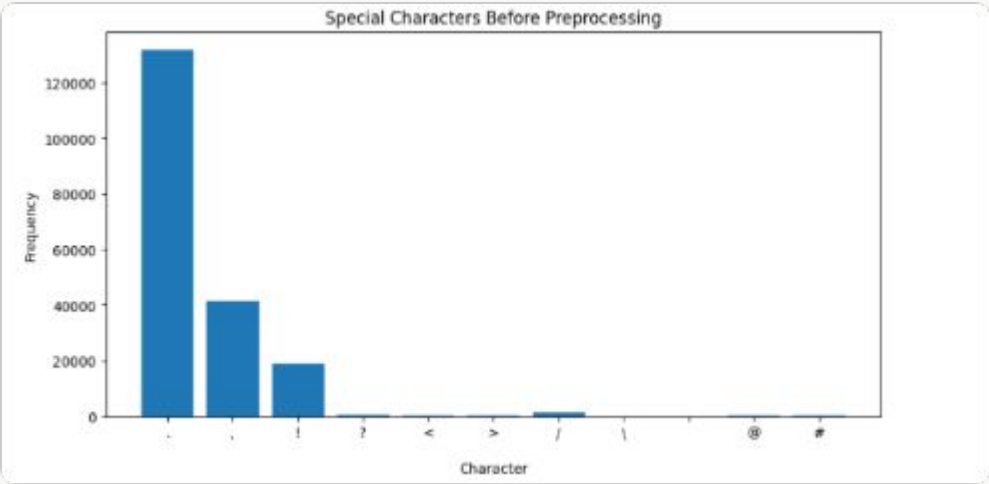
Built an end-to-end NLP pipeline combining preprocessing, sentiment classification, semantic clustering, and Generative AI summarization

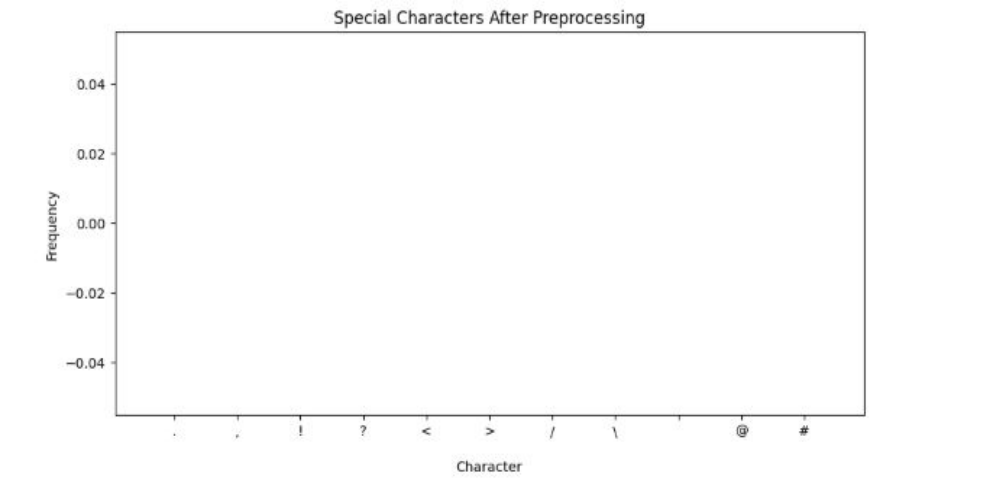
Dataset

Amazon Reviews dataset containing product reviews, ratings, product names, and customer feedback

Preprocessing

- Cleaned review text by removing special characters, line breaks, punctuation, sentences shorter than 10 letters and inconsistent formatting
- Standardized text for model input





Special Characters Before Preprocessing

Model 1: DistilBERT Sentiment Classifier

The Problem

- Customer reviews are highly unstructured and difficult to analyze manually at scale
- Companies need to automatically understand whether feedback is positive, neutral, or negative
- Manual analysis is time-consuming and inconsistent

The Solution

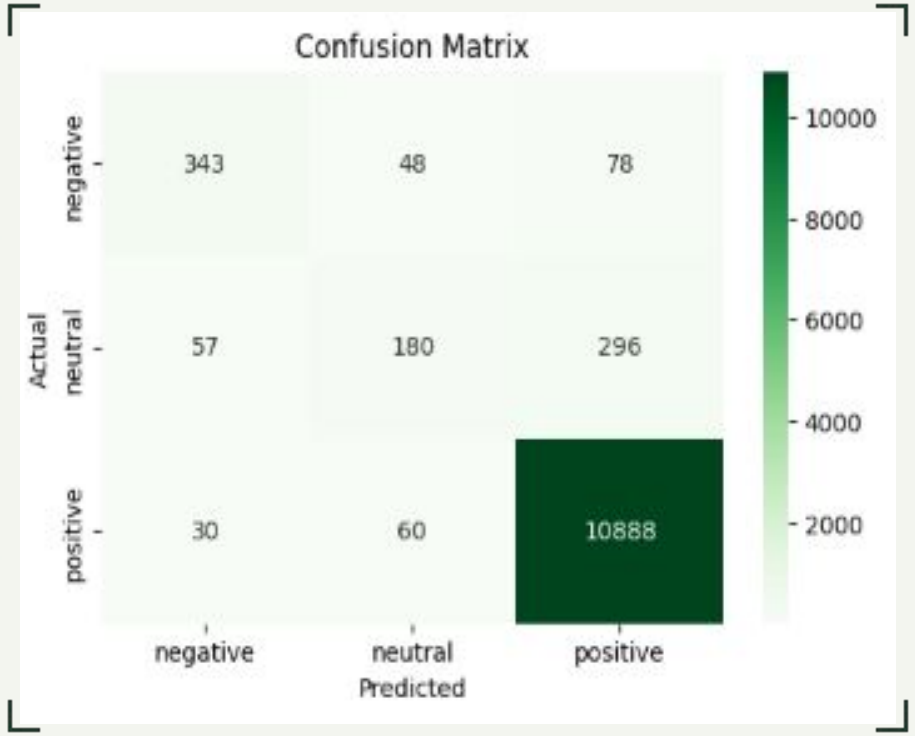
- Used DistilBERT, a transformer-based NLP model for sentiment classification
- Why DistilBERT? Lightweight, fast, and highly effective for text classification tasks
- Reviews were cleaned and labeled using review ratings, then used to train and evaluate the model

DistilBERT Performance Results

Results DistilBERT achieved strong classification performance with approximately 95% accuracy

Accuracy	Precision
0.953	0.946

Recall	F1-score
0.953	0.948



Why It Works

DistilBERT understands word context and sentence meaning, helping detect sentiment more accurately even in complex reviews

Evaluation Method

Evaluated using Accuracy, Precision, Recall, F1-score, and confusion matrix

Model 2: Product Clustering

The Problem

Too many products

Difficult to analyze individually

Cluster 3
batteries
great
price
last
work
good
brand
amazon
battery
long
better
remote
duracell
product
one
well
used
buy
time
seem
recommend
box
far
will
mouse
deal
best
value
energizer
aa
aaa
life
day
now
don
use
ame
ve
ught
uch
lashlights
battery
package
expensive
perfect
worked
month
less
even
sure
see
light
still
fine
going
came
cheap
quality
purchased
excellent
definitely
amazonbasic
expected
hour
basics
recommend
disappointed
four
longer
without
pay
buying
back
week
getting
store
right
toy
make
go
bulk
charge
christmas
three
using
couple
computer
shelf
got
short
put
home
thought
re
year
lasted
overall
replaced
replace
set
say
half
rice
rain
kids
usage
second
warning
easy
kind
quantity
bad batch
found
happens
hard
dollar store
order
paying
thing
cardboard box
type
never
didn
haven
electronic
super
yet
small
cost
need
worth
highly
name
item
cheaper
size
general
ll
worst
performance
every
standard
come
always
review
change
money
packaging
tv
waste
cycle
low
first
around
dollar
make
toy
go
bulk
charge
home
comfort
easy
usage
second
warning

Clustering Results

Results

Silhouette score is **relatively low (~0.07)**, indicating weak cluster separation. This is expected for text data, where reviews often share similar vocabulary

Evaluation Method

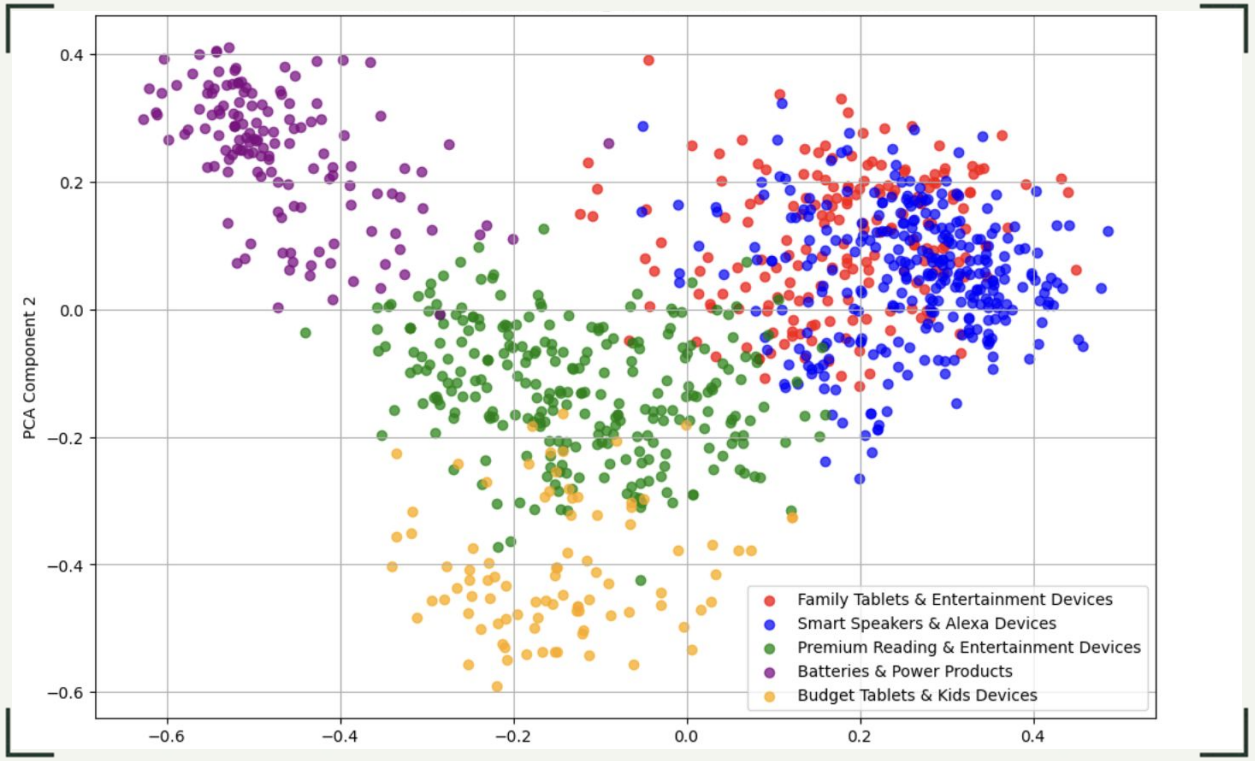
Visual inspection of clusters to assess semantic coherence and meaningful groupings.

Why It Works

Semantic clusters are clearly visible in the visualization, showing meaningful product groupings.

Cluster Names

Named manually according to the patterns observed in the cluster visualizations.



Model 3: AI-Generated Recommendations

The Problem

- Too many reviews to read manually
- Need automatic product recommendations per category
- Users want quick, actionable insights

The Solution

- Prompt engineering: Asked the model to summarize customer opinions
- Identify best products based on sentiment and clustering results
- Generate short, readable recommendation articles

Output

Concise articles with top products, common feedback, and buying recommendations

Generative AI Output

1

CLUSTER 1

Smart Speakers & Alexa Devices

TOP RATED PRODUCTS

Echo Plus with Built-In Hub (Silver) | Amazon Tap Smart Assistant Alexa-enabled Speaker | Echo (White)

ANALYSIS

Among Amazon smart speaker devices, the Echo Plus with Built-In Hub (Silver) achieved the highest performance with a 4.92 rating from 12 reviews. The Amazon Tap Smart Assistant Alexa-enabled speaker followed closely with a 4.75 rating, while the standard Echo (White) maintained strong customer satisfaction with a 4.72 rating across 36 reviews.

COMMON COMPLAINTS

- Voice recognition problems
- Inconsistent music playback
- Incorrect Alexa responses

WEAKEST PRODUCT — AVOID

Amazon Tap Portable Bluetooth Speaker (4.0 rating)

★★★★★ OUR RECOMMENDATION

For smart home usage and voice assistant functionality, the Echo Plus with Built-In Hub is the strongest recommendation in this cluster.

Evaluation Approach

Assessed quality of generated recommendation articles based on relevance, readability, and actionability.

Sample Output

AI-generated recommendation articles combining sentiment analysis, clustering insights, and product recommendations.

Key Features

- Concise summaries of customer opinions
- Top/Weakest products highlighted
- Common complaints identified
- Clear buying recommendations

Project Recap

ReviewSense AI successfully combines three AI models (DistilBERT, MiniLM+KMeans, Generative AI) into a complete review analysis pipeline

Final Version

End-to-end system that transforms 50,000+ unstructured reviews into actionable product recommendations

Key Takeaway

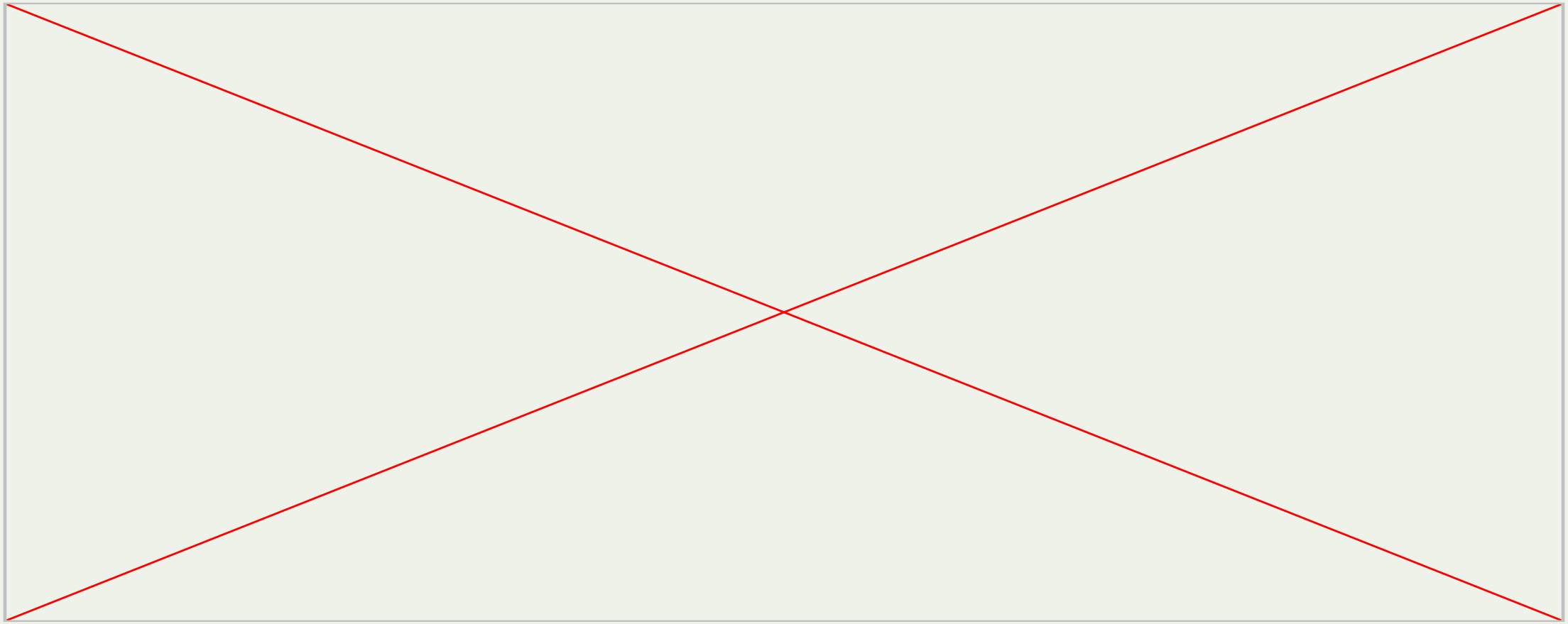
Multi-model NLP pipelines can automate complex analysis tasks and deliver real value to users

Bonus Point

The system is scalable and can be adapted to any product category or review dataset

From data chaos to clarity — that's the power of AI! ✨

Live Demo



What You'll See:

- Top products per cluster
- Our recommendation
- most usual complaints
- Final AI-generated recommendation article

Thank You

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Questions?